PATIENT CARE THEORY 2

UNIT 11, PART 3: Other Endocrine Disorders

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Winter 2020, 2021, 2023

Learning Outcomes

Upon successful completion of this unit, the student will have reliably demonstrated the ability to:

- Discuss the role of endocrine glands
- Discuss the role of hormones of the endocrine system
- List the common endocrine emergencies paramedics will encounter
- Compare endocrine emergencies in terms of sensitivity and specificity of signs and symptoms
- Contrast metabolic from structural cause of altered level of awareness
- Describe and defend your treatment strategy for endocrine emergencies

Endocrine System

- Includes 8 major glands
  - Each gland releases 1 or more hormones
  - Endocrine glands are ductless*
- Closely linked to the nervous system
- Uses hormones as chemical messengers

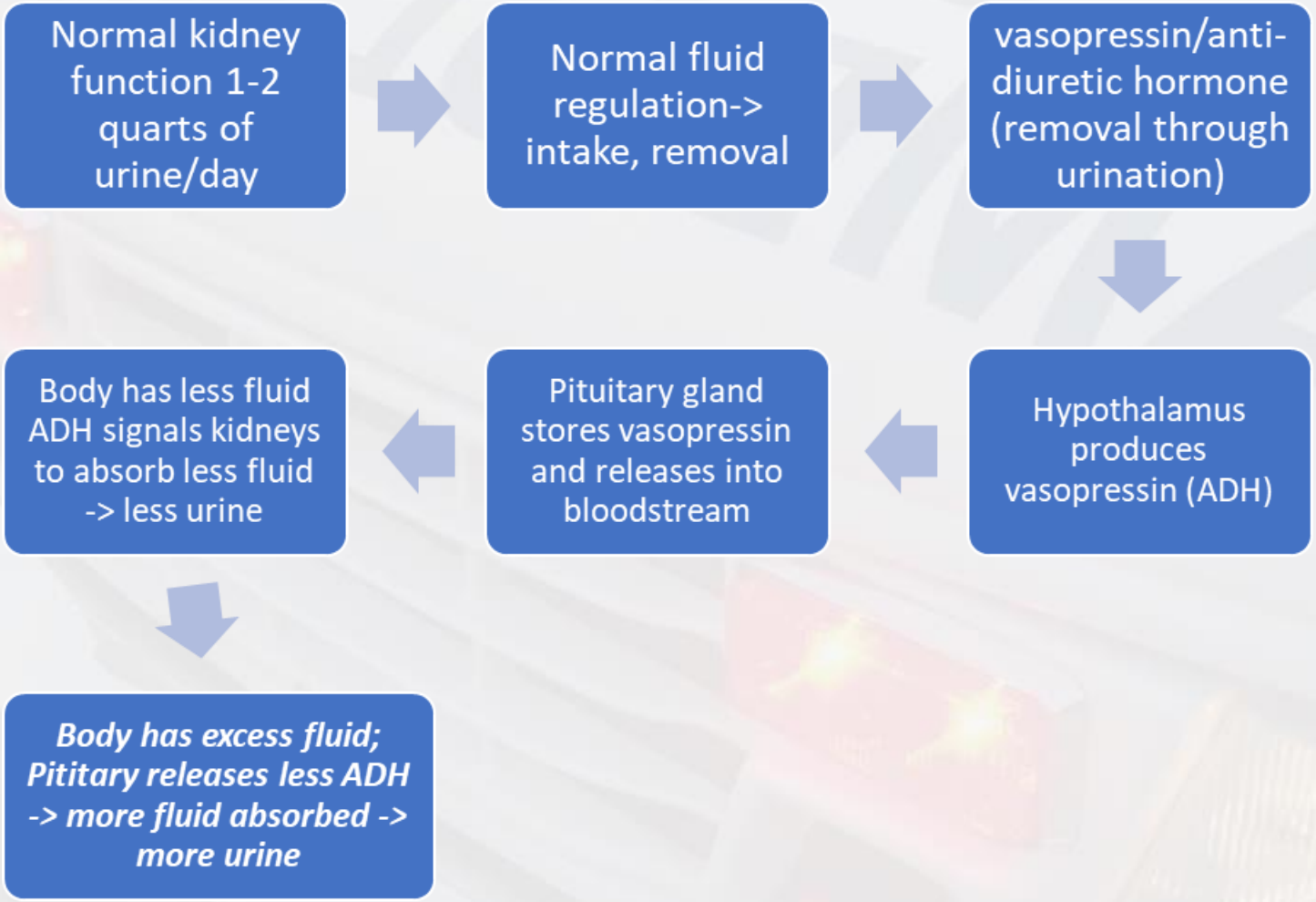
MAJOR ENDOCRINE GLANDS

- There are 8 major endocrine glands
  - Hypothalamus
  - Pituitary
  - Thyroid
  - Parathyroid
  - Thymus
  - Pancreas
  - Adrenals
  - Gonads (Ovaries/Testis)

Diabetes Insipidus

- Characterized by decreased secretion of antidiuretic hormone (ADH)
  - ADH controls the fluid removal rate through urination
  - Hypothalamus produces ADH (aka vasopressin)
  - Pituitary stores ADH and releases it into the bloodstream when low fluid volume is sensed
  - ADH signals kidneys to either absorb more or less fluid = more or less urine

Fluid regulation in the body



Diabetes Insipidus

- Characterized by reduced levels of ADH (vasopressin) to be released.
  - A hormone deficiency OR the kidneys reduced response to the hormone
  - Result in poor water retention leading to dehydration

DIABETES MELLITUS

VERSUS

DIABETES INSIPIDUS

Diabetes mellitus is a group of metabolic diseases characterized by excessive levels of the sugar glucose in the blood	Diabetes insipidus is a condition characterized by excessive thirst and excretion of large amounts of severely dilute urine
Commonly caused by a deficiency of the pancreatic hormone insulin, which results in a failure to metabolize sugars and starch	Commonly caused by a deficiency of the pituitary hormone vasopressin, which regulates kidney function
Urine contains glucose	Urine doesn't contain glucose

Pediaa.com

Diabetes Insipidus

- S&S:
  - Polyuria
  - Polydipsia
  - Nocturia (waking up 2 or more times at night to urinate)
- Severe dehydration, hypernatremia, fever, cardiovascular collapse, and death can occur in:
  - Children
  - elderly people
  - those with complicating illnesses

Diabetes Insipidus

Management

- Most people with diabetes insipidus replenish fluids well on their own
- In the event of a 9-1-1 call, look for signs of dehydration and hypotension and treat with IV NaCL prn

Disorders of the Adrenal Glands

- Addison's disease
- Cushing's Syndrome (not to be confused with Cushing's triad)

Adrenal Gland Disorders

- Addison's: characterized by adrenocortical insufficiency
  - Symptoms include: weakness, fatigue, weight loss, hyperpigmentation
- Cushing's Syndrome: pathological condition resulting from excess adrenocortical hormones
  - Symptoms include: changed body habitus, hypertension, increased susceptibility to infection

Addison's Disease (primary adrenal insufficiency)

- Autoimmune (or infectious, neoplastic, traumatic, vascular, etc.) ***destruction of the adrenal cortex***
- Symptoms appear when $\geq$ 90% is destroyed
- glucocorticoid (cortisol), mineralocorticoid (aldosterone), and androgen hormones are lacking
- Cortisol=stress hormone
- Major electrolyte balances
- Can be precipitated by steroid therapy (secondary adrenal insufficiency) called ***Addisonian Crisis***

Addison's Disease

- Nausea, vomiting, and diffuse abdominal pain are present in approximately 90% of patients and usually represent an impending Addisonian crisis.
  - Diarrhea is less common than nausea, vomiting, and abdominal pain and occurs in approximately 20% of patients.
- Chronic fatigue, progressive muscle weakness
- Anorexia and weight loss
- depression, irritability, and decreased concentration
- hyperpigmentation or dark tanning
  - most visible on scars; skin folds; pressure points such as the elbows, knees, knuckles, and toes; lips; and mucous membranes

Addison's Disease

Addison's Disease

9-1-1 call

- adrenal crisis; Addisonian crisis; acute adrenal insufficiency
- abrupt, life-threatening state caused by insufficient cortisol (cortisol is normally released in response to stress to ↑ BP and BGL)
  - headache, confusion
  - profound weakness, slow, sluggish, lethargic movement
  - nausea/Vomiting
  - Hypotension (syncopal episodes)
  - dehydration
  - high fever, chills shaking
  - tachycardia, tachypnea
  - abdominal pain – either diffuse of flank pain
  - unusual and excessive sweating on face and/or palms
  - shock, coma, seizure

Suspected Adrenal Crisis

Indications

A patient with primary adrenal failure who is experiencing clinical signs of an adrenal crisis.

Conditions

	Hydrocortisone
Age	N/A
LOA	N/A
HR	N/A
RR	N/A
SBP	N/A
Other	Paramedics are presented with a vial of hydrocortisone for the identified patient **AND** Age-related hypoglycemia **OR** GI symptoms (vomiting, diarrhea, abdominal pain) **OR** Syncope **OR** Temperature ≥38C or suspected/history of fever **OR** Altered level of awareness **OR** Age-related tachycardia **OR** Age-related hypotension

Suspected Adrenal Crisis

Contraindications

Hydrocortisone
Allergy or sensitivity to hydrocortisone

Treatment

Consider hydrocortisone	
	Route
	IM/IV
Dose	2 mg/kg*
Max. single dose	100 mg
Dosing interval	N/A
Max. # of doses	1

*Dose should be rounded to the nearest 10 mg

ALS PCS 5.3

Cushing's Syndrome

- Excess cortisol: caused by prolonged exposure to elevated levels of either endogenous glucocorticoids or exogenous glucocorticoids (Rx steroids)
  - Increased Na+ retention -> increased blood volume; together with increased sensitivity to catecholamines = hypertension
- Women more often than men
- Management of fats is altered by the body leading to increased incidence of hypercholesteremia and atherosclerosis
- Increased risk of DM
- Cortisol has anti-inflammatory and immunosuppressive properties which leads to increased susceptibility for infections

Cushing's Syndrome

- Physical manifestations
  - moon facies (Thickening of facial fat, which rounds the facial contour)
  - facial plethora (excess of blood in the facial veins, causing a ruddy complexion)
  - buffalo hump (increased accumulation of fat in the back)
  - supraclavicular fat pads
  - Skin changes (thinning, easily bruised, delayed healing)
  - truncal obesity
  - purple striae (stretch marks)
  - Mood swings
  - Increased development of facial hair (women)

Cushing's Syndrome

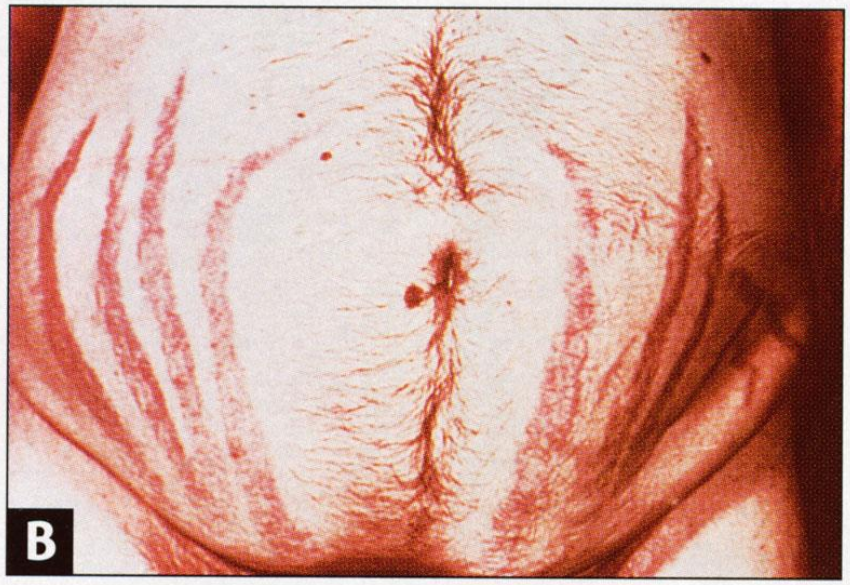
- moon facies
- facial plethora

Cushing's Syndrome

- buffalo hump

Cushing's Syndrome

- purple striae



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Cushing's Syndrome

Signs & Symptoms

- Hypertension
- Obesity
- Osteoporosis → predisposition for fractures
- impaired immune function → opportunistic infections
  - sepsis
- impaired wound healing
- glucose intolerance
- psychosis (e.g. depression, cognitive dysfunction, and emotional lability)

Hyperaldosteronism

- Patho: Aldosterone is produced and secreted by the adrenal glands
  - signals the kidneys to excrete less sodium and more potassium
  - Aldosterone production is regulated partly by corticotropin (secreted by the pituitary gland) and partly through the renin-angiotensin-aldosterone system

Hyperaldosteronism

- Causes HTN (5-15% of HTN)
- Unlike other cause of HTN, this can be cured with the proper surgical intervention
- overproduction of aldosterone leads to:
- HTN from fluid retention
- Hypokalemia
  - Weakness
  - Tingling
  - muscle spasms, and periods of temporary paralysis
  - Tachycardia
  - dysrhythmias

Disorders of the Thyroid Gland

- Hyperthyroidism
  - Goiters
  - Grave’s disease
- Thyrotoxicosis
- Hypothyroidism
- Myxedema

Role of the Thyroid

- Regulates tissue and body metabolism and development
- Thyroid hormones regulate protein synthesis
- Under control from the pituitary gland

9-1-1: hyperthyroidism is more likely to present acutely in the field than hypothyroidism

However, you'll likely encounter patients with a history of *hypo*thyroidism but who have called 9-1-1 for another reason unrelated to their thyroid disorder.

Goiter

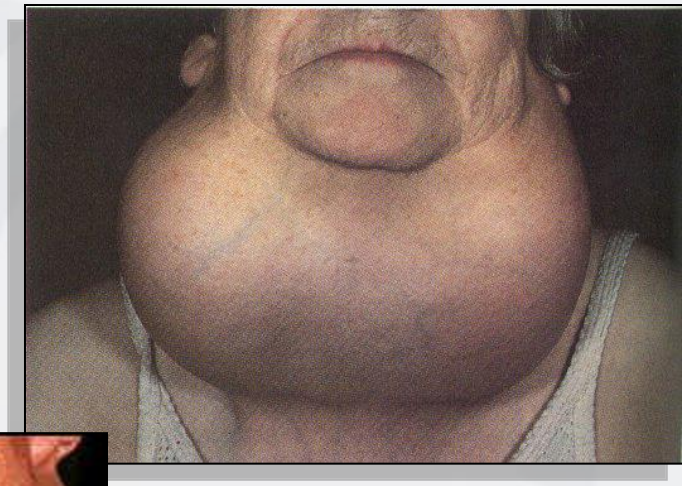
- An enlarged thyroid gland
  - may extend into the retrosternal space
- A deficiency in thyroid hormone synthesis or intake leads to increased TSH production.
- Causes: inborn errors of thyroid hormone synthesis, iodine deficiency, and goitrogens
- increased cell production and hyperplasia of the thyroid gland in an attempt to normalize thyroid hormone levels
- If this process is sustained, a goiter occurs
- most goiters are due to autoimmune thyroiditis (ie, Hashimoto disease.) or associated with Graves' Disease

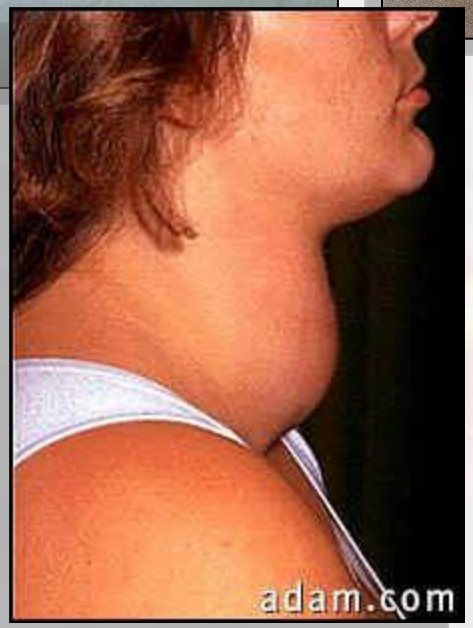
Goiter

- Enlargement of the thyroid gland may lead to compression of the:
  - trachea
  - larynx
  - superior and inferior laryngeal nerves
  - esophagus
- However, most goiters are benign, causing only cosmetic disfigurement

Goiter

example





Graves' Disease

- Accounts for >95% of thyrotoxicosis
- Genetic predisposition
- Autoimmune: autoantibodies are generated that produce excess thyroid hormones due to overstimulation of the thyroid tissue
- The excess thyroid hormone and/or autoantibody generation result in organ function changes

Hyperthyroidism - thyrotoxicosis

- excessive synthesis of thyroid hormones
- accelerated metabolism in the peripheral tissues
- leads to increased cardiac output at rest
  - but to decreased muscle bulk and function
  - Cardiac irritability
  - Hyper-response of beta-adrenergic receptors (i.e. ↑ sensitivity to catecholamines)
- Thyrotoxic Crisis, on the other hand, refers to the clinical effects of an unbound thyroid hormone, whether or not the thyroid gland is the primary source

Hyperthyroidism - thyrotoxicosis

- Signs & Symptoms
  - Confusion
  - dyspnea
  - Nervousness and tremor, emotional lability
  - Muscle weakness
  - Heat intolerance
  - Weight loss (with increased appetite)
  - Palpitations (e.g. paroxismal atrial fibrillation)

Hyperthyroidism – thyrotoxic crisis

- Thyroid Storm
  - Rare life-threatening condition due to hyperthyroidism
  - Seen most often in patients with Hx of Grave's Disease (most common cause of hyperthyroidism in children)
  - Usually precipitated by infection or some other stressful event
- S&S:
  - Related to enhance SNS activity
  - Spiked temperature (> 40° C)
  - Diaphoresis
  - Altered LOA: Confusion → delirium → seizures → coma
  - Tachycardia, other dysrhythmias (e.g. A. Fib., PVCs)
  - Clues to Dx: widened pulse pressure, palpable goiter

Hyperthyroidism - thyrotoxic crisis

- Management
  - Mainly supportive
  - O2, PPV prn
  - Fluid resuscitation
  - Cardiac monitoring
  - transport

Hypothyroidism

- Can be congenital or acquired
- Causes a low metabolic state due to low levels of thyroid hormones circulating in the blood
- Poor responses to exercise and/or infection

Hypothyroidism

- Weakness, fatigue, lethargy, and somnolence
- Cold intolerance, decreased sweating
- Dry, coarse skin
- Headache
- Swelling of the face and extremities
- Impaired memory and cognition, poor concentration
- Mild weight gain (with anorexia)
- Coarseness of voice and impaired hearing
- Paresthesias and arthralgias
- Muscle cramps
- Constipation

Hypothyroidism - Myxedema

- Myxedema is a rare, life-threatening manifestation of severe hypothyroidism
- Seen most often in the winter months in women with undiagnosed hypothyroidism
- Signs
  - Dry scaly yellow skin, non-pitting waxy edema of the face and extremities (myxedema) and thinning eyebrows.
  - CVS: bradycardia, enlarged heart, low voltage ECG
  - Paresthesia
  - Ataxia
  - **Myxedema coma**: altered LOA, hypothermia, hypoglycemia, hypotension, bradycardia, bradypnea, electrolyte imbalances, delusions, psychosis

Myxedema

Thyroiditis

- Inflammation of the thyroid gland
- Causes: bacterial or viral infection, or autoimmune
- Hashimoto's thyroiditis (autoimmune disorder) is a chronic disorder and is the most common cause of hypothyroidism
- Thyrotoxicosis can also occur with thyroiditis

Hypo/hyperthyroidism - management

- Supportive care
  - SpO_2 monitor
  - O_2 prn
  - PPV prn
  - **ECG** monitor (especially important with hyperthyroidism)
  - IV TKO
  - Transport

Summary

- Majority of emergencies related to the endocrine system will involve complications associated with diabetes mellitus
  - Hypoglycemia
  - Hyperglycemia
    - DKA
    - HHNS

Additional Reading

Addison's Disease
http://www.emedicine.com/derm/topic761.htm

Cushing's Syndrome
http://www.emedicine.com/emerg/topic117.htm

Diabetes Insipidus
http://www.emedicine.com/med/topic543.htm

Hyperaldosteronism
http://www.emedicine.com/med/topic3193.htm

Hyperthyroidism
http://www.emedicine.com/PED/topic1099.htm

Hypothyroidism
http://www.emedicine.com/med/topic1145.htm

QUESTIONS?